# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 1 – Data Interpretation

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| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO4 – Precision | **Assessment:** | Assessment Tool 1 – Data Interpretation |
| **Weightage:** | 40% of CIA | | |
| **CO5 Statement:** | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) | | |
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| **Section / Batch:** | | **Date:** | |

**Code of Conduct**

*I __C. Anu Keerthana(192472057)________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___C. Anu Keerthana______________________

**Instructions**

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

| Section / Topic | Focus | Question No. |
|---|---|---|
| Representing Meaning & Meaning Structure of Language | Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems. | Q1 |
| First-Order Predicate Calculus & Representing Linguistically Relevant Concepts | Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments. | Q2 |
| Lexemes and Their Senses & Word Sense Disambiguation | Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems. | Q3 |

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|---|---|---|
| Syntax-Driven Semantic Analysis | Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications. | Q4 |

## Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

| Customer Query | Semantic Representation |
|---|---|
| "Activate international roaming for my number." | ACTIVATE(Roaming, Customer) |
| "Deactivate caller tune service." | DEACTIVATE(CallerTune, Customer) |
| "Check my data balance." | QUERY(DataBalance, Customer) |
| "Enable 5G service." | ACTIVATE(5GService, Customer) |

Additional chatbot logs show:

| Query ID | Actual User Intent | Predicted Intent |
|---|---|---|
| Q1 | Activate Roaming | Activate Roaming |
| Q2 | Deactivate Caller Tune | Activate Caller Tune |
| Q3 | Check Data Balance | Query Data Balance |
| Q4 | Enable 5G Service | Activate 5G Service |

**Tasks:**

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules. Production Data:

| Machine | Status |
|---|---|
| M1 | Active |
| M2 | Active |
| M3 | Maintenance |
| M4 | Active |

**Predicate Rules:**

- Active(x) → Producing(x)
- Produces(x,y)∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

**Tasks:**

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

| User Query | Possible Sense 1 | Possible Sense 2 |
|---|---|---|
| "Apple accessories" | Fruit | Technology Brand |
| "Mouse wireless" | Animal | Computer Device |
| "Java tutorial" | Island | Programming Language |
| "Python course" | Snake | Programming Language |

Search Context Data:

| Query | Clicked Result |
|---|---|
| Apple accessories | iPhone Charger |
| Mouse wireless | Bluetooth Mouse |
| Java tutorial | Coding Lessons |
| Python course | Software Development Training |

**Tasks:**

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

| Sentence | Parse Structure |
|---|---|
| Doctor prescribed medicine to patient. | Subject-Verb-Object |
| Patient reported severe headache. | Subject-Verb-Object |
| Nurse monitored patient continuously. | Subject-Verb-Object |
| Medicine reduced blood pressure. | Subject-Verb-Object |

Semantic Roles Generated:

| Entity | Role |
|---|---|
| Doctor | Agent |
| Medicine | Instrument |
| Patient | Recipient |
| Headache | Symptom |

**Tasks:**

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.

4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

## Q1. Semantic Representation in Customer Support Chatbots

**1. Analyze the semantic representations and identify the action-object relationships.**

The semantic representations follow the general structure:

**ACTION (Object, Entity)**

where the first argument represents the object or service being acted upon and the second argument represents the entity associated with the action, which is the customer.

| Customer Query | Semantic Representation | Action | Object |
|---|---|---|---|
| Activate international roaming for my number. | ACTIVATE(Roaming, Customer) | ACTIVATE | Roaming |
| Deactivate caller tune service. | DEACTIVATE(CallerTune, Customer) | DEACTIVATE | Caller Tune |
| Check my data balance. | QUERY(DataBalance, Customer) | QUERY | Data Balance |
| Enable 5G service. | ACTIVATE(5GService, Customer) | ACTIVATE | 5G Service |

The representations correctly capture the main intent of each query. For example, **ACTIVATE(Roaming, Customer)** indicates that the customer wants to activate the roaming service, while **QUERY(DataBalance, Customer)** indicates that the customer wants information about the available data balance.

Thus, the semantic representation separates the **customer's intended action** from the **service or information involved**, allowing the chatbot to map natural-language requests to appropriate system operations.

**2. Determine which query contains semantic interpretation errors.**

The error occurs in **Q2**.

- **Actual intent:** Deactivate Caller Tune
- **Predicted intent:** Activate Caller Tune

The correct semantic representation should be:

**DEACTIVATE(CallerTune, Customer)**

However, the chatbot incorrectly interprets the request as:

**ACTIVATE(CallerTune, Customer)**

This is a significant semantic error because **activate** and **deactivate** represent opposite actions. If the chatbot executes the predicted intent, it may enable a service that the customer specifically requested to disable.

The other three queries have matching actual and predicted intents.

**3. Evaluate how meaning representation affects chatbot decision accuracy.**

Meaning representation directly affects the accuracy of chatbot decision-making because it converts a user's natural-language request into a structured form that can be processed by the system.

For example:

**"Deactivate caller tune service."**

should become:

**DEACTIVATE(CallerTune, Customer)**

If it is incorrectly represented as:

**ACTIVATE(CallerTune, Customer)**

the chatbot may perform the wrong operation.

From the given data, **3 out of 4 queries are correctly interpreted**, giving an observed intent accuracy of:

**Accuracy = 3/4 × 100 = 75%**

Therefore, the chatbot achieves **75% accuracy** on the given sample, while Q2 demonstrates that an error in semantic interpretation can directly lead to an incorrect customer-service action.

**4. Recommend improvements to enhance semantic understanding.**

The following improvements can increase semantic understanding:

1. **Intent classification:** Train the system to distinguish opposite intents such as activate/deactivate, enable/disable, and subscribe/unsubscribe.
2. **Entity recognition:** Identify important entities such as roaming, caller tune, data balance, and 5G service.
3. **Context analysis:** Consider previous conversation turns before determining the user's intent.
4. **Semantic validation:** Check whether the predicted action is consistent with the user's words.
5. **Synonym handling:** Recognize expressions such as "turn off," "disable," and "deactivate" as related intents.

6. **Confidence scoring:** If the model has low confidence, the chatbot should ask the customer for confirmation instead of executing the operation.

**Conclusion**

The given semantic representations effectively capture the relationship between customer actions and telecom services. However, Q2 demonstrates that confusing opposite actions can produce incorrect decisions. Therefore, accurate intent classification, entity recognition, context analysis, and semantic validation are essential for reliable telecom chatbots.

## Q2. First-Order Predicate Calculus for Smart Manufacturing

**1. Represent the production data using First-Order Predicate Calculus.**

The machine-status information can be represented using predicates.

Given:

- M1 is Active
- M2 is Active
- M3 is under Maintenance
- M4 is Active

The corresponding predicates are:

**Active(M1)**
**Active(M2)**
**Maintenance(M3)**
**Active(M4)**

The rules provided are:

**Rule 1:**
Active(x) → Producing(x)

**Rule 2:**
Produces(x,y) ∧ Active(x) → Available(y)

**Rule 3:**
Maintenance(x) → ¬Producing(x)

Using Rule 1:

- Active(M1) → Producing(M1)
- Active(M2) → Producing(M2)
- Active(M4) → Producing(M4)

Using Rule 3:

- Maintenance(M3) → ¬Producing(M3)

Therefore:

Producing(M1)
Producing(M2)

¬Producing(M3)
Producing(M4)

**2. Analyze which products are currently available.**

According to Rule 2:

**Produces(x,y) ∧ Active(x) → Available(y)**

This means a product is considered available when it is produced by an active machine.

The data confirms that M1, M2, and M4 are active. Therefore, any products produced by these machines can be considered available.

However, the question does **not provide an explicit mapping between machines and products**. Therefore, specific product names cannot be logically identified from the supplied data.

The correct conclusion based strictly on the given information is:

- Products produced by **M1, M2, and M4** can be inferred as available.
- Products that depend only on **M3** cannot be considered currently produced because M3 is under maintenance.

**3. Infer whether Gear production is affected by maintenance activities.**

The available information does not explicitly state which machine produces **Gear**.

If Gear is produced by M3, then:

**Maintenance(M3) → ¬Producing(M3)**

Therefore:

**Gear production is affected/stopped.**

If Gear is produced by M1, M2, or M4, its production is not affected by M3's maintenance.

Hence, based strictly on the given predicates, **Gear production can only be concluded to be affected if Gear is produced by M3**.

This distinction is important because FOPC allows conclusions only when the required facts and relationships are available.

**4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.**

First-Order Predicate Calculus is useful in smart manufacturing because it represents machine states and operational rules formally.

For example:

**Active(M1) → Producing(M1)**

allows the system to automatically infer that M1 is producing.

Similarly:

**Maintenance(M3) → ¬Producing(M3)**

allows the system to identify that M3 is not producing.

Advantages include:

- Formal and unambiguous representation.
- Automatic logical inference.
- Easy representation of machine conditions.
- Support for rule-based decision-making.
- Ability to identify operational consequences.

However, the system depends on the availability and accuracy of facts. For example, the given data does not specify which machine produces Gear. Therefore, the system cannot safely infer Gear's status without that relationship.

**Conclusion**

FOPC provides a structured framework for representing machine states and deriving production-related conclusions. In this scenario, M1, M2, and M4 are producing, while M3 is not producing because it is under maintenance. However, product-specific conclusions require additional machine-product relationships.

## Q3. Word Sense Disambiguation in E-Commerce Search Engines

**1. Analyze the contextual information and determine the correct word sense for each query.**

The correct senses can be determined from the clicked search results.

| Query | Possible Senses | Clicked Result | Correct Sense |
|---|---|---|---|
| Apple accessories | Fruit / Technology Brand | iPhone Charger | Technology/Brand |
| Mouse wireless | Animal / Computer Device | Bluetooth Mouse | Computer Device |
| Java tutorial | Island / Programming Language | Coding Lessons | Programming Language |
| Python course | Snake / Programming Language | Software Development Training | Programming Language |

Therefore:

1. Apple accessories → Apple technology brand
2. Mouse wireless → Computer mouse
3. Java tutorial → Java programming language
4. Python course → Python programming language

**2. Identify the semantic cues used for disambiguation.**

The major semantic cues are:

**a. Query words**

Words surrounding the ambiguous term provide useful information.

For example:

**"accessories"** suggests a product or technology context rather than a fruit.

**"tutorial"** suggests educational content, and in this case the clicked result is related to programming.

**"course"** and **"software development training"** strongly indicate that Python refers to the programming language.

**b. Clicked search result**

The clicked result provides strong evidence about the user's intended meaning.

For example:

**Apple accessories → iPhone Charger**

indicates that Apple refers to the technology brand.

**c. Domain context**

The fact that the system is an e-commerce/search environment also helps distinguish product-related meanings from unrelated meanings.

**3. Evaluate the impact of incorrect sense selection on search engine performance.**

Incorrect word-sense selection can significantly reduce search quality.

For example, if **"Python course"** is interpreted as a snake-related query, the search engine may return:

- Snake biology courses
- Reptile information
- Wildlife-related products

instead of:

- Python programming courses
- Programming tutorials
- Software development training

Similarly, interpreting **"Apple accessories"** as fruit could produce irrelevant food-related results.

This can lead to:

- Lower search relevance.
- Reduced click-through rate.
- Poor recommendations.
- Increased user frustration.
- Lower customer satisfaction.
- Potential loss of sales in e-commerce applications.

Therefore, WSD is important for connecting a user's query with the correct semantic meaning.

**4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.**

An industrial-scale WSD system can use a combination of methods:

1. **Contextual embeddings:** Use transformer-based language models to understand words based on surrounding context.
2. **Query classification:** Classify queries according to domains such as technology, food, programming, or animals.
3. **User behavior:** Use clicks, previous searches, purchases, and browsing history when appropriate.
4. **Knowledge bases:** Connect ambiguous words with structured entities and concepts.
5. **Search-result feedback:** Use clicked results as evidence for the intended sense.
6. **Continuous learning:** Update the model using new queries and user interactions.
7. **Hybrid WSD:** Combine machine-learning predictions with rules and domain-specific knowledge.

**Conclusion**

The search logs demonstrate that context is essential for resolving lexical ambiguity. The words surrounding the ambiguous term and the clicked results provide strong semantic evidence. Accurate WSD improves search relevance and recommendation accuracy, while incorrect sense selection can produce irrelevant results and negatively affect user experience.

## Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

**1. Analyze how syntactic structures contribute to semantic role assignment.**

Syntactic structure establishes relationships between the entities and actions in a sentence.

For example:

**"Doctor prescribed medicine to patient."**

The basic structure is:

**Subject → Verb → Object**

- Subject: Doctor
- Verb: Prescribed
- Object: Medicine
- Recipient: Patient

This syntactic information allows the semantic system to assign roles:

- Doctor → Agent

- Medicine → Instrument/Object
- Patient → Recipient

Similarly:

**"Patient reported severe headache."**

can be analyzed as:

- Patient → Experiencer
- Headache → Symptom

And:

**"Nurse monitored patient continuously."**

can be interpreted as:

- Nurse → Agent
- Patient → Object/Patient being monitored

Therefore, syntax provides the structural information required for assigning semantic roles.

**2. Determine whether the assigned semantic roles are appropriate.**

The provided roles are:

| Entity | Assigned Role | Evaluation |
|---|---|---|
| Doctor | Agent | Appropriate |
| Medicine | Instrument | Reasonable in the given representation |
| Patient | Recipient | Appropriate for "prescribed medicine to patient" |
| Headache | Symptom | Appropriate |

The **Doctor → Agent** relationship is correct because the doctor performs the action of prescribing.

The **Patient → Recipient** relationship is also appropriate because the patient receives the prescribed medicine.

**Headache → Symptom** is appropriate because it represents the patient's reported medical symptom.

The role **Medicine → Instrument** can be used in a broad semantic representation, although it is more precisely the **object/theme of prescribing** in the sentence "Doctor prescribed medicine to patient."

Thus, most roles are appropriate, but the medicine role should be carefully defined according to the semantic-role framework being used.

**3. Evaluate potential errors that could arise from incorrect parsing.**

Incorrect syntactic parsing can result in incorrect semantic interpretation.

For example, if the system incorrectly identifies **patient** as the subject instead of the recipient in:

**"Doctor prescribed medicine to patient."**

the system may infer that the patient performed the prescribing action.

This could produce serious errors such as:

- Incorrect identification of the responsible medical professional.
- Incorrect assignment of patient information.
- Wrong medication relationships.
- Incorrect extraction of symptoms and treatments.
- Errors in medical records.
- Incorrect clinical decision support.

In healthcare applications, such errors are particularly important because semantic information may be used for patient records and medical decision-making.

**4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.**

The following methods can improve the system:

1. **Medical-domain parsers:** Use NLP models trained specifically on medical text.
2. **Dependency parsing:** Identify grammatical relationships between medical entities.
3. **Named Entity Recognition:** Detect doctors, patients, medicines, symptoms, diseases, and treatments.
4. **Semantic Role Labeling:** Assign roles such as Agent, Patient, Recipient, Instrument, and Symptom.
5. **Domain-specific ontologies:** Use medical knowledge bases to improve semantic interpretation.
6. **Context-aware models:** Consider surrounding sentences when assigning roles.
7. **Human validation:** Important medical information should be reviewed when model confidence is low.
8. **Error analysis:** Regularly evaluate the parser using annotated medical datasets.

## Conclusion

Syntax provides the structural foundation for semantic interpretation. Correct parsing helps the system identify who performed an action, what was affected, and who received treatment. Because healthcare information is sensitive and clinically important, syntax-driven semantic analysis should combine accurate parsing, medical-domain knowledge, semantic role labeling, and validation mechanisms.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Data | 25% | Accurately interprets all data patterns, trends, and relationships | Interprets data correctly with minor gaps | Partial understanding; misses key trends | Misinterprets or fails to understand data |
| Analysis | 25% | Provides deep analysis with strong logical reasoning and justification | Provides reasonable analysis with some justification | Basic analysis with weak reasoning | No meaningful analysis provided |
| Application of Concepts | 20% | Effectively applies relevant concepts/tools to interpret data accurately | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply concepts to data |
| Conclusion | 20% | Draws clear, meaningful conclusions with strong insights and implications | Provides valid conclusions with moderate insights | Conclusions are vague or weak | No clear or valid conclusions |
| Clarity | 10% | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues | Lacks clarity or proper structure | Poor presentation and difficult to understand |

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
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